

# PSet 2

1) C

2) a)  $\begin{bmatrix} 9 \\ 12 \\ 3 \end{bmatrix} \rightarrow \begin{matrix} wx \\ wy \\ w \end{matrix}$

↓

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{9}{3} \\ \frac{12}{3} \end{bmatrix} = \boxed{(3, 4)}$$

b)  $\begin{bmatrix} 9 \\ 12 \\ 3 \end{bmatrix} \rightarrow \boxed{(9, 12, 3, 1)}$   
append a 1

3) A

4) a) x-axis

b) B

c) B

d) C

e) C

5) a) Extrinsic matrix:  $E = [R|t]$

| camera axis | world axis: | camera-real world offset:                        |
|-------------|-------------|--------------------------------------------------|
| $x_c$       | $x_w$       | $C = \begin{bmatrix} 0 \\ -5 \\ 4 \end{bmatrix}$ |
| $y_c$       | $-z_w$      |                                                  |
| $z_c$       | $y_w$       |                                                  |

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$t = -RC = - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -5 \\ 4 \end{bmatrix} = - \begin{bmatrix} 0 \\ -4 \\ -5 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 5 \end{bmatrix}$$

$$\boxed{E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -4 \\ 0 & 1 & 0 & -5 \end{bmatrix}}$$

b)  ${}^c p = R \cdot {}^w p + t$

$$R \cdot {}^w p = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ -5 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ -5 \end{bmatrix}$$

$${}^c p = \begin{bmatrix} 3 \\ -1 \\ -5 \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \\ 5 \end{bmatrix} = \boxed{\begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix}}$$

c)  $\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \cdot \begin{bmatrix} {}^c p \\ z_c \end{bmatrix}, z_c = 1$

$$K = \begin{bmatrix} 100 & 0 & 200 \\ 0 & 100 & 100 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\boxed{(u, v) = (300, 300)}$$

$${}^c p = \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix} \Rightarrow K \cdot {}^c p = \begin{bmatrix} 300 \\ 300 \\ 0 \end{bmatrix}$$

b) a) c

b) b

c) c

d) b

e)  $z=5.2, \sigma=1$

$$A: \text{Likelihood} = e^{\left(\frac{-(5.2-4)^2}{2}\right)} \approx 0.49$$

$$B: \text{Likelihood} = e^{\left(\frac{-(5.2-5)^2}{2}\right)} \approx 0.98$$

$$C: \text{Likelihood} = e^{\left(\frac{-(5.2-6)^2}{2}\right)} \approx 0.73$$

Unnormalized weights:

$$A: 0.3 \cdot 0.49 \approx 0.147$$

$$B: 0.4 \cdot 0.98 \approx 0.392$$

$$C: 0.3 \cdot 0.73 \approx 0.219$$

Normalized:

$$\sum \text{unnormalized weights} = 0.758$$

$$A: \frac{0.147}{0.758} \approx 0.194$$

$$B: \frac{0.392}{0.758} \approx 0.517$$

$$C: \frac{0.219}{0.758} \approx 0.289$$